

A young man with short, curly brown hair, wearing a dark blue t-shirt, is seen from the side, looking out of a large glass window. His right hand is resting on the window frame. In the foreground, there are large, green, out-of-focus leaves of a plant, possibly a palm or similar tropical plant, which partially obscure the view. The background outside the window shows a bright, sunny day with some greenery and a blue metal railing visible in the distance. The overall lighting is warm and natural, suggesting an indoor setting with a view of the outdoors.

Input og Scanner

Program

- Casting
- Parsing
- Scanner
- System.out og System.in
- Øvelser
- Opsamling



Casting



```
int x = 4;  
double y = x; // går godt, y er nu 4.0
```

... men ikke den modsatte vej

```
double x = 2.7;  
int y = x; // fejl
```

... vi kan dog *explicit* konvertere

```
double x = 2.7;  
int y = (int) x; // går godt, - men...
```

Spørgsmål: Hvad er x?

```
char x = 'A'  
int y = (int) x; // går godt, c er nu 65
```

Hvad med **String**s?

```
String x = "4";  
int y = s; // fejl
```




Nyt ord: **Parsing**.

at fortolke en tekststreng (eller data)

```
String x = "4";  
int y = Integer.parseInt(s); // går godt, x er nu 4
```

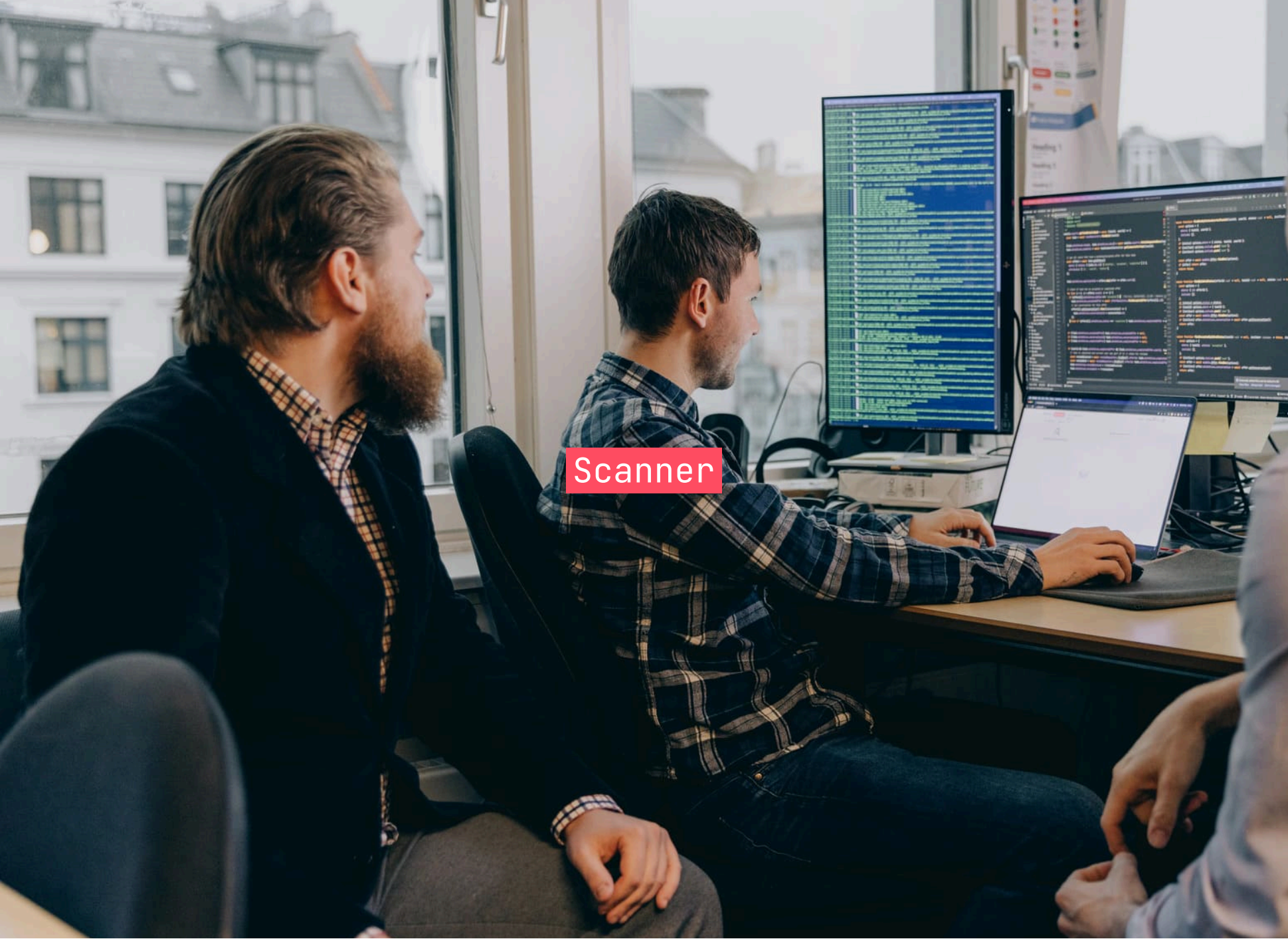
```
int x      = Integer.parseInt("4");  
double y   = Double.parseDouble("2.7");  
boolean z  = Boolean.parseBoolean("true");
```

```
int x = Integer.parseInt("4 121 33 14"); // fejl
```

... **parseInt** kan kun håndtere én enkelt værdi ad gangen.

Løsning: Brug **Scanner** klassen!

```
String s = "4 121 33 14";  
  
Scanner scanner = new Scanner(s);  
  
int x = scanner.nextInt(); // læser 4  
int y = scanner.nextInt(); // læser 121  
int z = scanner.nextInt(); // læser 33  
int w = scanner.nextInt(); // læser 14
```

Scanner

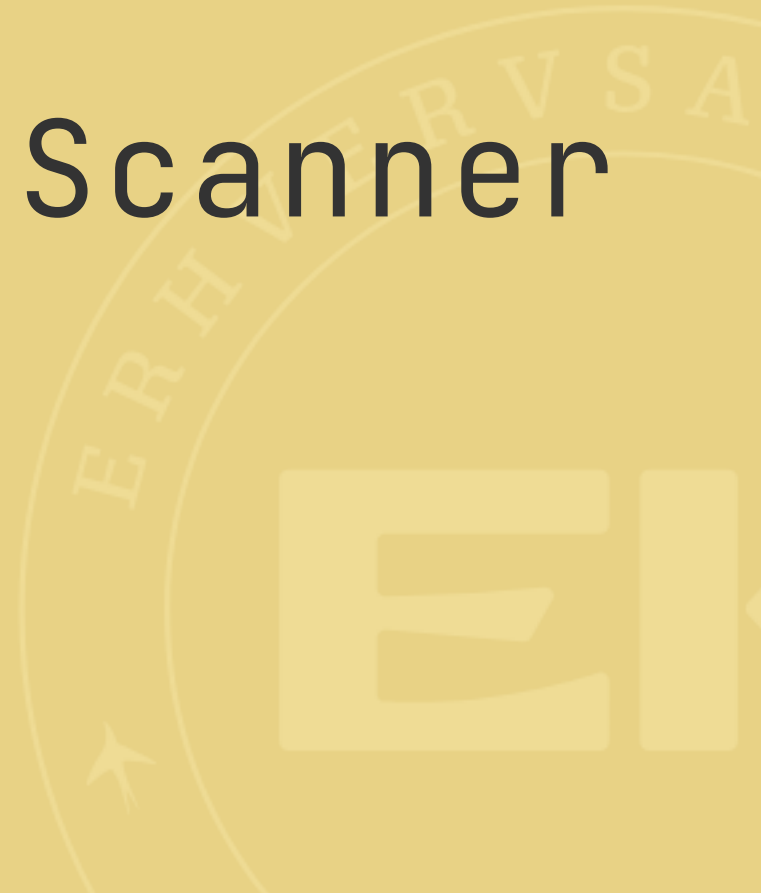
Scanner.java:

```
import java.util.Scanner;
public class Scanner {
    public String nextLine() {
        // ...
    }

    public int nextInt() {
        // ...
    }

    public double nextDouble() {
        // ...
    }
}
```

Demo: Klassen Scanner



A close-up photograph of a young woman with long, wavy blonde hair. She is smiling and looking off to the right. She is wearing a blue and white horizontally striped sweater. The background is blurred, showing what appears to be a brick building and some greenery.

System.out og System.in

System.out

- Vi har brugt `System.out.println` til at udskrive tekst til skærmen en del gange.
- Hvad er `System.out`?

```
System.out.println(System.out); // java.io.PrintStream@15db9742
```

HelloWorld.java:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

System.java:

```
public class System {  
    public static final PrintStream out = new PrintStream(...);  
    // ...  
}
```

PrintStream.java:

```
public class PrintStream {  
    public void println(String s) {  
        // ...  
    }  
  
    public void print(String s) {  
        // ...  
    }  
  
    // ...  
}
```

System.in

System.java:

```
public class System {  
    public static final InputStream in = new InputStream(...);  
    // ...  
}
```

InputStream.java:

```
public class InputStream {  
    public int read() {  
        // ...  
    }  
}
```

En **int**? GYS!

Demo af System.ins read
metode

Scanner-klassen **to the rescue!**

Demo af Scanner og System.in

A close-up photograph of a young woman with long, wavy blonde hair, smiling and looking off to the side. She is wearing a blue and white horizontally striped sweater. The background is a blurred outdoor setting with a brick building and some greenery.

Scanner s hasNext -
metoder

Vi kan ikke forvente at brugeren altid indtaster det rigtige. Hvordan undgår vi fejl, når vi bruger **Scanner**?

Demo - fejlintastning

En måde er at bruge **hasNext**-metoderne:

```
Scanner scanner = new Scanner(System.in);
int x = 0;
if (scanner.hasNextInt()) {
    x = scanner.nextInt();
    System.out.println("Du indtastede: " + x);
} else {
    System.out.println("Fejl: Det indtastede er ikke et tal.");
}
```

Er der flere tal i inputtet?

```
Scanner scanner = new Scanner(System.in);  
while (scanner.hasNextInt()) {  
    int x = scanner.nextInt();  
    System.out.println("Du indtastede: " + x);  
}
```


Er der flere ord i inputtet?

```
Scanner scanner = new Scanner(System.in);  
while (scanner.hasNext()) {  
    String word = scanner.next();  
    System.out.println("Du indtastede: " + word);  
}
```

Er der flere linjer i inputtet?

```
Scanner scanner = new Scanner(System.in);  
while (scanner.hasNextLine()) {  
    String line = scanner.nextLine();  
    System.out.println("Du indtastede: " + line);  
}
```

Demo - hasNext...

A young man with dark, curly hair and a goatee is smiling while wearing large black headphones. He is sitting outdoors, wearing a blue t-shirt, and typing on a silver laptop. The background is a vibrant, out-of-focus urban scene with graffiti on a wall and green plants. A white rectangular box is superimposed over the middle of the image, containing the text 'Refaktorisering af kode' in a black serif font.

Refaktorisering af kode

De to faser i kodning

1. **Løs problemet** - få noget kode til at virke - **slamkode**
2. **Ryd op** - gør den mere læsbar og vedligeholdelsesvenlig - **refaktorisering**

Slamkode

Refaktorisering - **Don't Repeat Yourself - DRY**